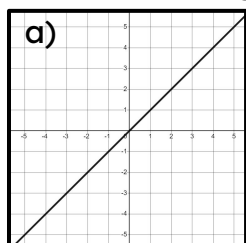




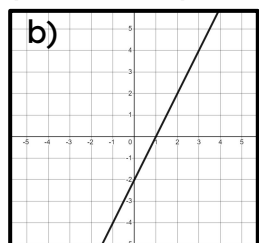
Finding the equation of the line $ay + bx + c = 0$

Task A: Equations in the form $y = mx + c$

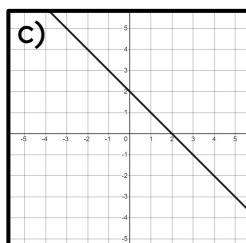
1) Match each graph to its equation



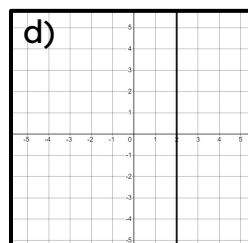
$y = x$



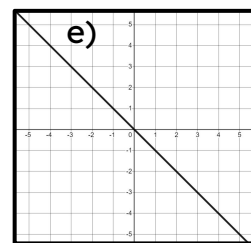
$y = 2x - 2$



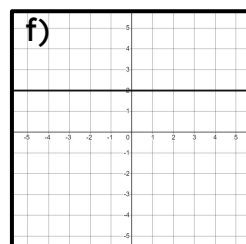
$y = 2 - x$



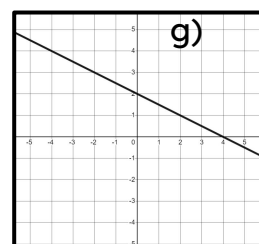
$x = 2$



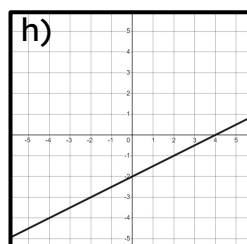
$y = -x$



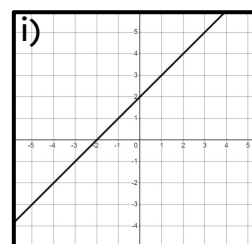
$y = 2$



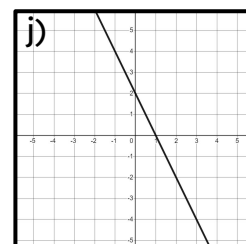
$y = 2 - \frac{1}{2}x$



$y = \frac{1}{2}x - 2$

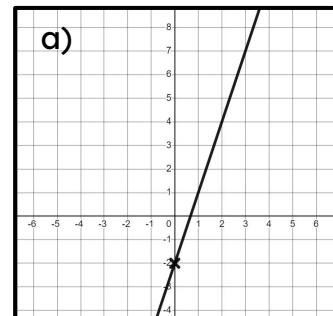


$y = x + 2$

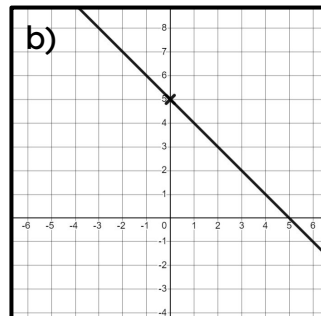


$y = 2 - 2x$

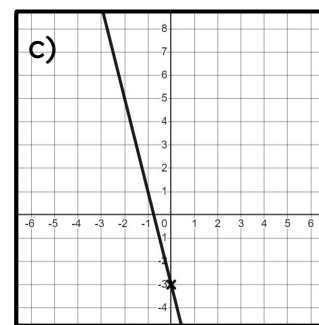
2) For each question, work out an equation for the line and use the key features to draw the line on the axes.



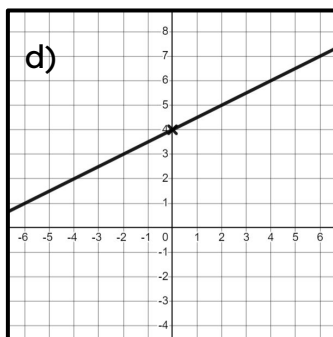
gradient: 3
y-intercept: (0, -2)
equation: $y = 3x - 2$



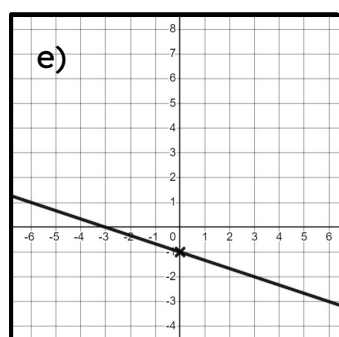
gradient: -1
y-intercept: (0, 5)
equation: $y = -x + 5$



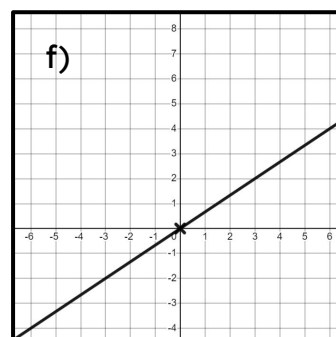
gradient: -4
y-intercept: (0, -3)
equation: $y = -4x - 3$



gradient: $\frac{1}{2}$
y-intercept: (0, 4)
equation: $y = \frac{1}{2}x + 4$



gradient: $-\frac{1}{3}$
y-intercept: (0, -1)
equation: $y = -\frac{1}{3}x - 1$



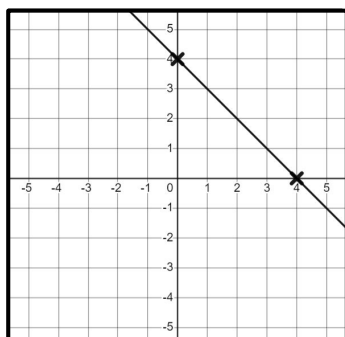
gradient: $\frac{2}{3}$
y-intercept: (0, 0)
equation: $y = \frac{2}{3}x$



Task B: Equations in the form $ay + bx + c = 0$

1) For each equation, work out the y -intercept, the x -intercept and draw the graph.

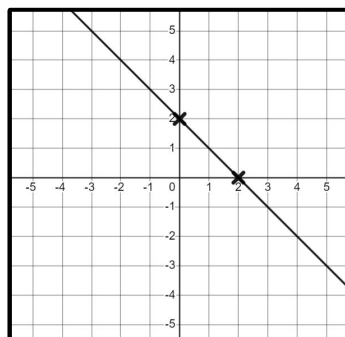
a) $y + x - 4 = 0$



y -intercept $(0, 4)$

x -intercept $(4, 0)$

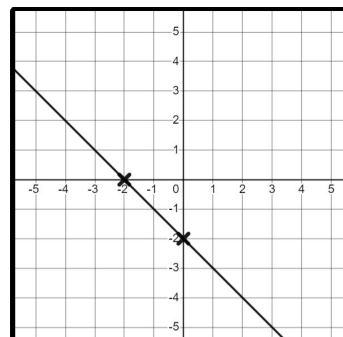
b) $y + x - 2 = 0$



y -intercept $(0, 2)$

x -intercept $(2, 0)$

c) $y + x + 2 = 0$



y -intercept $(0, -2)$

x -intercept $(-2, 0)$

2) From the bar model, write the relationship as an equation in the form $y = mx + c$ and $ax + by + c = 0$

a)

y		2
x	x	x

$y = mx + c : y = 3x - 2$

$ax + by + c = 0 : 3x - y - 2 = 0$

b)

10		
y	x	x

$y = mx + c : y = 10 - 2x$

$ax + by + c = 0 : 2x + y - 10 = 0$

3) Fill in the missing working, then draw the graph of $2y + 5x - 20 = 0$

y -intercept

$2y + 5(0) - 20 = 0$

$2y - 20 = 0$

$2y = 20$

$y = 10$ $(0, 10)$

x -intercept

$2(0) + 5x - 20 = 0$

$5x - 20 = 0$

$5x = 20$

$x = 4$ $(4, 0)$

